Reinforcement Learning III
Value Functions

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What will be covered

- What are Value Functions?
- Bellman Equations
- Action Value Functions

Recall: MDP

$$< S, A, T, R, \boldsymbol{\gamma} >$$

- S: Set of states i.e. encode environment configurations
- A: Set of actions i.e. the decisions made by the agent
- T: Transition function i.e. how the environment evolves under actions

$$T(s, a, s') = P(s_{t+1} = s' | s_t = s, a_t = a)$$

- R: Reward function i.e. immediate feedback signal $R(s, a) = c$
- $\gamma \in (0, 1]$: discounting for future rewards

Beyond MDPs

- What about
  - Policies?
  - Returns?
  - Value functions?
- Derived from the MDP

Return G

- Cumulative discounted return G represents the total accumulated reward an agent receives from a particular point in time onwards.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{H} \gamma^k R_{t+k+1}$$

The farther in the future a reward is, the less I care about it, exponentially.

- R_{t+1} represents the immediate reward
- What is H?
- Importance of the return G
  - Defines what agents aim to maximise in RL
  - Used in the definition of value function in MDPs.

Policy π

- Policy π represents the agent's behaviour
- Mapping from states to actions $a = \pi(s)$
  - Can be stochastic: $\pi(a|s) = P[a_t = a|s_t = s]$
  - $\sum_{a \in A} \pi(a \mid s) = 1 \; \forall \, s \in S$
- Why?
  - To encourage exploration
  - Optimality and robustness in uncertain environments

Policy π

- Goal of RL is to find an optimal policy π that maximises the expected discounted return

$$J(\pi) = \mathbb{E}_{\pi}[G_0] = \mathbb{E}[\sum_{t=0}^{\infty}(\gamma^t R\,(s_t\,, a_t\,)|\,\pi]$$

A policy tells the agent what to do in any situation it might encounter, guiding its behaviour to achieve its goals.

Value Functions

Value Functions

- How valuable is it to be in state s while following π?

Each action contributes to the state's value in proportion to the total future rewards it leads to.

$$V(s) = \mathbb{E}[return\ from\ state\ s]$$
$$= \mathbb{E}_{\boldsymbol{a\sim\pi(\cdot|s),s'\sim P(\cdot|s,a)}}[\boldsymbol{R(s,a) + \gamma V^{\pi}(s')}]$$

*Equation assumes that both the policy and the environment are stochastic (averaging over outcomes weighted by probability).

Average over the policy's possible actions and the environment's possible next states.

- $V^{\pi}(s) = R(s, \pi(s)) + \gamma V^{\pi}(s')$ // fully deterministic version

Value Functions

- How valuable is it to be in state s while following π?



*The value function is not always bounded between 0 and 1 like this example. Its range depends on the reward scale and discount factor.

Model-based Value Functions

Give an action credit in proportion to all of the rewards that followed it

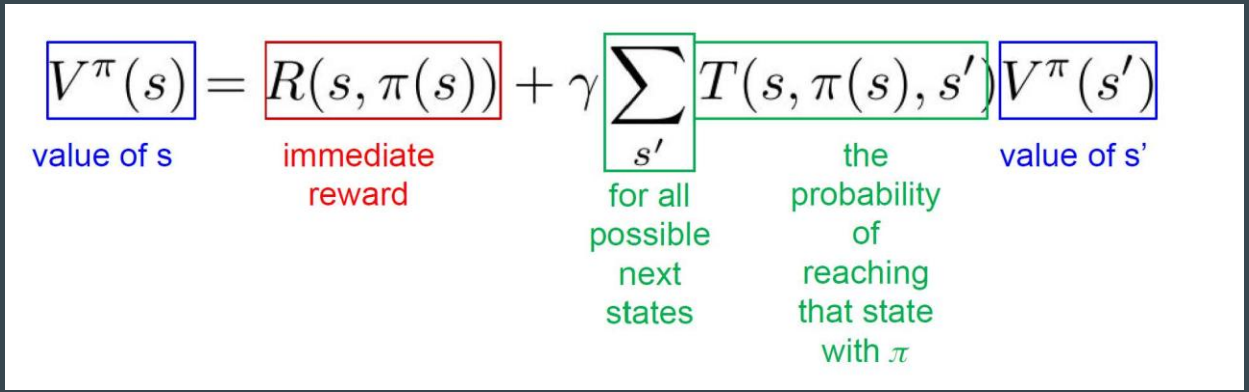


Figure: Bellman Equation

Bellman Equations

- Recursively defines the expected return
- Can compute V or Q using Bellman!

Derivation of the Bellman Equation

Derivation of $V^{\pi}(s) = \mathbb{E}_{a\sim\pi, s'\sim P}[R(s,a) + \gamma V^{\pi}(s')]$

- By definition: $V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$ where $G_t = \sum_{k=0}^{H} \gamma^k R_{t+k+1}$
- $V^{\pi}(s) = \mathbb{E}_{\pi}[\, R_{t+1} + \gamma G_{t+1} \mid S_t = s\,]$ --($G_t = R_{t+1} + \gamma G_{t+1}$)
- $V^{\pi}(s) = \mathbb{E}_{\pi}[R_{t+1} \mid S_t = s] + \gamma \mathbb{E}_{\pi}[G_{t+1} \mid S_{t+1} = s']$ --(split expectation)
- $V^{\pi}(s) = \mathbb{E}_{\pi}[R(s,a) + \gamma V^{\pi}(s')]$ --($\mathbb{E}_{\pi}[G_{t+1} \mid S_{t+1} = s'] = V^{\pi}(s')$)

Current value = immediate reward + γ × future value

Action Value Functions

- How good is it to take a specific action a in state s if using policy π for all subsequent decisions?

$$Q(s,a) = E[\text{return after taking action } a \text{ in state } s]$$
$$= \mathbb{E}_{\boldsymbol{s' \sim P(\cdot|s,a)}}\left[\boldsymbol{R(s,a)} + \boldsymbol{\gamma}\mathbb{E}_{\boldsymbol{a' \sim \pi(\cdot|s)}}[\boldsymbol{Q^{\pi}(s',a')}])\right]$$

Average over the policy's possible actions and the environment's possible next states.

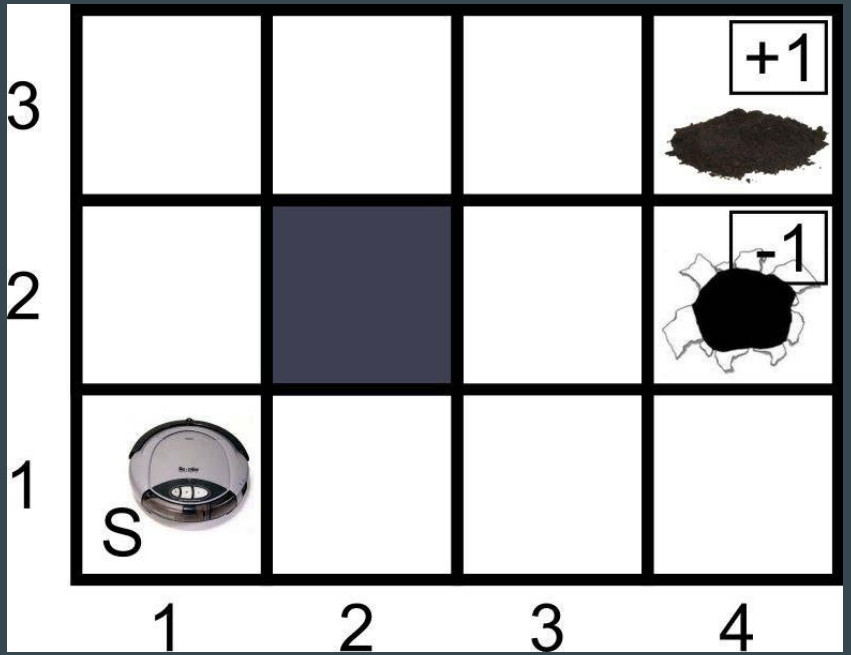
- Represents the "quality" of taking action a in state s
- Equation assumes that both the policy and the environment are stochastic

Exercise: Derive the recursive equation on your own!
$Q^{\pi}(s,a) = R(s,a) + \gamma\, Q^{\pi}(s',a')$ //fully deterministic version

Relationship Between V and Q

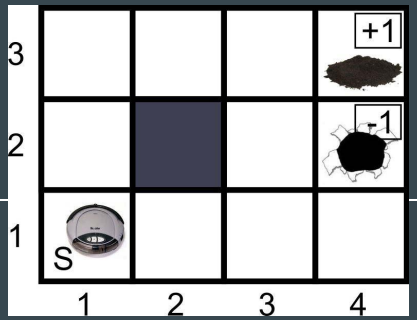
- $V^{\pi}(s) = \mathbb{E}_{a\sim\pi}[Q^{\pi}(s,a)]$
- The value of a state is simply the average value of all the actions you could take there, weighted by how likely your policy is to pick each action.
- $Q^{\pi}(s,a) = \mathbb{E}_{s\prime\sim P}[R(s,a,s') + \gamma V\pi(s')]$
- Do I want to evaluate states or actions?

Recall: Gridworld



- States: Position on grid
  - S is (1,1), goal (4,3)
  - OR grid of values
- Actions: {Up, Down, Left, Right}
- Reward:
  - +1 for finding dirt
  - -1 for falling into hole
  - -0.001 for every move
  - Why those specific values?

Example: Gridworld



Given: $\gamma = 0.9, \pi(Right \mid (3,3)) = 0.8, \pi(Down \mid (3,3)) = 0.2, \pi(Left \mid (3,3)) = 0$
Assume environment is deterministic

$Q^{\pi}(s,a) = R(s,a) + \gamma V^{\pi}(s')$
$Q^{\pi}\big((3,3), Right\big) = R\big((3,3), Right\big) + \gamma V^{\pi}(3,4)$
$= 0.999 + 0.9 \cdot 0 = 0.999$ // Goal is terminal $\therefore V^{\pi}((3,4)) = 0$

$Q^{\pi}\big((3,3), Down\big) = R\big((3,3), Down\big) + \gamma V^{\pi}(2,3)$
$= -0.001 + 0.9 \cdot 0 = -0.001$ // Using current estimate $V^{\pi}(2,3) = 0$ from initialisation
Similarly, $Q^{\pi}\big((3,3), Left\big) = -0{,}001$
Up is not a legal action in $(3,3)$ so it is not computed

$V^{\pi}(s) = \sum_{a \in legal\ A} \pi(a \mid s) Q^{\pi}(s,a)$
$V^{\pi}(3,3) = \pi(Right \mid (3,3)) + Q^{\pi}\big((3,3), Right\big) + \pi(Down \mid (3,3)) + Q^{\pi}\big((3,3), Down\big) + \pi(Left \mid (3,3)) Q^{\pi}((3,3), Left)$
$= 0.8(0.999) + 0.2(-0.001)) + 0\ (-0.001) = 0.799$